

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{2}{81} \sum_p g_1^4 LF_{3,0}[m_d^P] \delta_{i1i2} + \frac{5}{108} \sum_p g_1^4 LF_{4,-1}[m_d^P] \delta_{i1i2} - \frac{8}{405} \sum_p g_1^4 LF_{5,-2}[m_d^P] \delta_{i1i2} - \right. \\ & \quad \frac{2}{27} \sum_p g_1^4 LF_{3,0}[m_e^P] \delta_{i1i2} + \frac{5}{36} \sum_p g_1^4 LF_{4,-1}[m_e^P] \delta_{i1i2} - \frac{8}{135} \sum_p g_1^4 LF_{5,-2}[m_e^P] \delta_{i1i2} - \\ & \quad \frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_l^P] \delta_{i1i2} + \frac{5}{72} \sum_p g_1^4 LF_{4,-1}[m_l^P] \delta_{i1i2} - \frac{4}{135} \sum_p g_1^4 LF_{5,-2}[m_l^P] \delta_{i1i2} - \\ & \quad \frac{1}{81} \sum_p g_1^4 LF_{3,0}[m_q^P] \delta_{i1i2} + \frac{5}{216} \sum_p g_1^4 LF_{4,-1}[m_q^P] \delta_{i1i2} - \frac{4}{405} \sum_p g_1^4 LF_{5,-2}[m_q^P] \delta_{i1i2} - \\ & \quad \frac{8}{81} \sum_p g_1^4 LF_{3,0}[m_u^P] \delta_{i1i2} + \frac{5}{27} \sum_p g_1^4 LF_{4,-1}[m_u^P] \delta_{i1i2} - \frac{32}{405} \sum_p g_1^4 LF_{5,-2}[m_u^P] \delta_{i1i2} - \\ & \quad \frac{1}{27} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} - \frac{1}{18} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} + \frac{8}{135} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} - \\ & \quad \frac{1}{324} g_1^4 LF_{2,1,0}[m_1, m_d^{i1}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{2,2,-1}[m_1, m_d^{i1}] \delta_{i1i2} + \\ & \quad \frac{1}{162} g_1^4 LF_{3,1,-1}[m_1, m_d^{i1}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{4,1,-2}[m_1, m_d^{i1}] \delta_{i1i2} - \\ & \quad \frac{1}{324} g_1^4 LF_{2,1,0}[m_1, m_d^{i2}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{2,2,-1}[m_1, m_d^{i2}] \delta_{i1i2} + \\ & \quad \frac{1}{162} g_1^4 LF_{3,1,-1}[m_1, m_d^{i2}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{4,1,-2}[m_1, m_d^{i2}] \delta_{i1i2} - \\ & \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i1}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i1}] \delta_{i1i2} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i1}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i1}] \delta_{i1i2} - \\ & \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i2}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i2}] \delta_{i1i2} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i2}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i2}] \delta_{i1i2} - \\ & \quad \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{2,2,0}[m_d^r, m_q^P] \delta_{i1i2} + \frac{1}{18} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{3,2,-1}[m_d^r, m_q^P] \delta_{i1i2} + \\ & \quad \frac{1}{162} g_1^4 LF_{2,1,0}[m_d^{i1}, m_1] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{3,1,-1}[m_d^{i1}, m_1] \delta_{i1i2} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i1}, m_3] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i1}, m_3] \delta_{i1i2} + \\ & \quad \frac{1}{162} g_1^4 LF_{2,1,0}[m_d^{i2}, m_1] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{3,1,-1}[m_d^{i2}, m_1] \delta_{i1i2} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i2}, m_3] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i2}, m_3] \delta_{i1i2} - \\ & \quad \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{2,2,0}[m_e^r, m_l^P] \delta_{i1i2} + \frac{1}{18} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{3,2,-1}[m_e^r, m_l^P] \delta_{i1i2} + \\ & \quad \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{3,1,0}[m_l^P, m_e^r] \delta_{i1i2} + \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{3,2,-1}[m_l^P, m_e^r] \delta_{i1i2} - \\ & \quad \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{4,1,-1}[m_l^P, m_e^r] \delta_{i1i2} + \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{5,1,-2}[m_l^P, m_e^r] \delta_{i1i2} + \\ & \quad \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{3,1,0}[m_q^P, m_d^r] \delta_{i1i2} + \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{3,2,-1}[m_q^P, m_d^r] \delta_{i1i2} - \\ & \quad \frac{5}{12} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{4,1,-1}[m_q^P, m_d^r] \delta_{i1i2} + \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{5,1,-2}[m_q^P, m_d^r] \delta_{i1i2} + \\ & \quad \frac{1}{18} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{2,2,0}[m_q^P, m_u^r] \delta_{i1i2} - \frac{1}{36} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{3,2,-1}[m_q^P, m_u^r] \delta_{i1i2} - \\ & \quad \frac{1}{18} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,0}[m_q^P, \tilde{\mu}] + \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,2,-1}[m_q^P, \tilde{\mu}] + \\ & \quad \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{3,1,-1}[m_q^P, \tilde{\mu}] - \frac{1}{18} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{3,1,0}[m_u^r, m_q^P] \delta_{i1i2} - \\ & \quad \frac{1}{18} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{3,2,-1}[m_u^r, m_q^P] \delta_{i1i2} - \frac{1}{6} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{4,1,-1}[m_u^r, m_q^P] \delta_{i1i2} + \\ & \quad \frac{1}{3} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{5,1,-2}[m_u^r, m_q^P] \delta_{i1i2} + \frac{5}{36} g_1^4 LF_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} - \\ & \quad \frac{1}{9} g_1^4 LF_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{5}{12} g_1^2 g_2^2 LF_{4,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{3} g_1^2 g_2^2 LF_{5,1,-3}[\tilde{\mu}, m_2] \delta_{i1i2} + \\ & \quad \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,0}[\tilde{\mu}, m_q^P] + \frac{7}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{3,1,-1}[\tilde{\mu}, m_q^P] - \\ & \quad \frac{1}{18} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{4,1,-2}[\tilde{\mu}, m_q^P] + \frac{1}{72} g_1^4 LF_{2,1,1,-1}[m_1, m_d^{i1}, \tilde{\mu}] \delta_{i1i2} + \\ & \quad \frac{1}{36} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_d^{i1}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{72} g_1^4 LF_{2,1,1,-1}[m_1, m_d^{i2}, \tilde{\mu}] \delta_{i1i2} + \\ & \quad \frac{1}{36} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_d^{i2}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{72} g_1^4 LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_d^{i1}] \delta_{i1i2} - \\ & \quad \frac{1}{72} g_1^4 m_1^2 LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_d^{i1}] \delta_{i1i2} + \frac{1}{72} g_1^4 LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_d^{i2}] \delta_{i1i2} - \\ & \quad \frac{1}{72} g_1^4 m_1^2 LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_d^{i2}] \delta_{i1i2} - \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_q^P] - \\ & \quad \frac{3}{8} g_2^2 \tilde{\mu}^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^P] + \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_q^P] + \\ & \quad \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline$$